
Causal Inference with Latent Markov Confounding under Imperfect Regime Overlap

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Abstract

Evaluating continuous policy interventions in observational atmospheric and environmental systems is fundamentally challenged by non-stationary confounding, temporal autocorrelation, and latent state ambiguity. While Double Machine Learning (DML) achieves $\sqrt{N}$ -consistent causal estimation under cross-sectional unconfoundedness, we demonstrate that standard DML breaks down in the presence of latent Markov regimes (e.g., planetary boundary-layer stagnation vs. advective ventilation). Crucially, we demonstrate that latent-regime causal estimation faces *two distinct, orthogonal failure modes*: (i) **Proxy Uncertainty** (ε_γ), governing latent state misclassification, and (ii) **Causal Overlap / Ill-Conditioning** ($\lambda_{\min}(\mathbf{J})$), governing collinearity and variance explosion in the coupled regime Gram matrix. We prove that predictive latent state inference alone cannot identify regime-specific causal effects under unconstrained overlap.

To address these challenges, we introduce **Overlap-Aware Regime Double Machine Learning (OR-DML)**. OR-DML establishes a rigorous **Graceful Degradation Bound**, proving that causal estimation error satisfies $\|\hat{\boldsymbol{\theta}}_\gamma - \boldsymbol{\theta}^*\|_2 \leq \frac{C\varepsilon_\gamma}{\lambda_{\min}(\mathbf{J})} + \mathcal{O}_P(N^{-1/2})$. To stabilize estimation under poor causal overlap, OR-DML introduces a **spectral regularization scheme** $\hat{\boldsymbol{\theta}}_\lambda = (\hat{\mathbf{J}} + \lambda\mathbf{I})^{-1}\hat{\mathbf{S}}$ establishing a finite-sample risk tradeoff, paired with **Purged Block-Temporal Cross-Fitting** that eliminates temporal leakage under α -mixing. In a 500-replication benchmark span-

ning separation regimes $\Delta_Z \in [0.2, 4.0]$, we chart the *Latent-Confounding Difficulty Frontier*, showing that while standard DML exhibits +8.44 bias and 0% coverage, OR-DML achieves nominal 94% coverage as separation increases. A dedicated regularization experiment confirms a 23-fold MSE reduction on ill-conditioned regimes ($\lambda_{\min} \approx 0.025$). Applied to $> 14,000$ verified hourly observations across four major Indian megacities (Delhi, Mumbai, Bengaluru, Kolkata), OR-DML resolves spurious confounding, shows that posterior entropy does not imply causal conditioning (demonstrated empirically in Kolkata), and maps dynamic causal impulse-response functions up to 24 hours.

1 INTRODUCTION

Estimating the causal impact of continuous interventions—such as curtailing vehicular emissions or industrial activity—on ambient air quality (PM_{2.5}) is critical for environmental policy (Dominici et al., 2014). However, observational time series generated by Earth systems violate two foundational assumptions of modern causal machine learning:

1. **Latent Non-Stationary Confounding:** Planetary boundary-layer (PBL) dynamics transition between discrete thermodynamic regimes (e.g., severe nocturnal inversion vs. advective convective clearance). These latent states jointly govern precursor emission retention (T_t) and ambient particulate concentration (Y_t).
2. **Imperfect Regime Overlap & Temporal Dependence:** Latent states are never directly observed; auxiliary meteorological proxies Z_t exhibit substantial overlap, making posterior state recovery uncertain ($\varepsilon_\gamma > 0$). Furthermore, autoregressive persistence breaks cross-sectional exchangeability (Chernozhukov et al., 2018), causing nuisance errors to leak across temporal folds.

When standard Double Machine Learning (Chernozhukov et al., 2018) or tree-based plug-in estimators are applied naively to environmental time series without accounting for latent regimes, the resulting causal estimands are severely biased, frequently suffering *sign inversions*—falsely suggesting that emission abatement exacerbates ambient pollution.

Related Work. Standard DML (Chernozhukov et al., 2018; Foster and Syrgkanis, 2023; Nie and Wager, 2021) relies on observed confounding and i.i.d. exchangeability. In time-series DML, Ciganovic et al. (2026) adapt DML to macroeconomic series via Reverse Cross-Fitting; however, their framework requires strict stationarity and time-reversibility, precluding non-stationary, regime-switching physical systems. In latent-state causal inference, Clouth et al. (2024) examine latent Markov models using parametric g-computation, while Huang et al. (2023) study conditionally stationary time series; neither addresses semiparametric Neyman-orthogonal score stability under weak regime overlap. Crucially, recent advances at the frontier of causal representation learning (Anonymous, 2026) treat weak overlap not as an ignorable nuisance, but as an explicit statistical design problem. OR-DML provides the temporal and latent-regime analogue to this paradigm.

Contributions. In this paper, we develop this framework:

- **Non-Identification under Unconstrained Overlap (Theorem 3):** We prove an impossibility result: without auxiliary separation or bounded proxy error, observational distributions with identical observables can arise from conflicting regime causal effects $\theta^{(1)} \neq \theta^{(2)}$. Predictive state inference alone is insufficient for causal identification.
- **Graceful Degradation Error Bound (Theorem 4):** Under explicit Lipschitz and moment regularity conditions, we derive the exact finite-proxy-error bound: $\|\hat{\theta}_\gamma - \theta^*\|_2 \leq \frac{C \cdot \varepsilon_\gamma}{\lambda_{\min}(\mathbf{J})} + \mathcal{O}_P(N^{-1/2})$, where ε_γ is the L_1 proxy error and $\lambda_{\min}(\mathbf{J})$ is the spectral floor of the coupled Gram matrix.
- **Spectral Regularization Bias-Variance Frontier (Theorem 5):** We establish the deterministic error identity $\theta_\lambda - \theta^* = (\mathbf{J} + \lambda \mathbf{I})^{-1}(\mathbf{b}_\gamma - \lambda \theta^*)$ and non-asymptotic MSE risk bound for the regularized estimator $\hat{\theta}_\lambda = (\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \hat{\mathbf{S}}$.
- **Omitted Regime Bias Decomposition (Theorem 1):** We derive the closed-form bias of naive DML, showing how residualizing treatment against controls shrinks $\mathbb{E}[\hat{T}_t^2]$, producing hyperbolic *Frisch-Waugh bias amplification*.

- **Dependent Data Asymptotics with Bartlett Embargo (Theorem 6):** We establish $\sqrt{N}$ -consistency and asymptotic normality under geometrically α -mixing processes via Purged Block Cross-Fitting with an empirical Bartlett embargo buffer τ^* .
- **The Latent-Confounding Difficulty Frontier (500 Replications):** Across five separation regimes $\Delta_Z \in [0.2, 4.0]$, we chart the transition from near-unidentifiability to clean separation, tracking $\lambda_{\min}(\mathbf{J})$, condition number $\kappa(\mathbf{J})$, and posterior entropy $\bar{H}(\gamma)$, and demonstrate nominal 94% coverage on both Sample and Population ATE targets.
- **Multi-City Longitudinal Empirical Study (14,000 Hours):** We evaluate OR-DML on clean, physically-bounded multi-season data across Delhi, Mumbai, Bengaluru, and Kolkata, demonstrating that posterior entropy does not equal causal conditioning, and mapping dynamic causal impulse responses up to 24 hours.

2 STRUCTURAL CAUSAL MODEL & IDENTIFICATION

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Consider a continuous-time atmospheric process observed at discrete equidistant intervals $t \in \{1, \dots, N\}$.

2.1 Structural Equations

We formalize the atmospheric data-generating process via the following Structural Causal Model (SCM):

$$S_t \sim \text{Markov}(\Pi, \pi_0), \quad S_t \in \{1, \dots, K\}, \quad (1)$$

$$Z_t = \mu_Z(S_t) + \xi_t, \quad (2)$$

$$X_t = \phi(X_{t-1}, S_t) + \epsilon_t, \quad (3)$$

$$T_t = m(X_t, S_t) + V_t, \quad \mathbb{E}[V_t | X_t, S_t] = 0, \quad (4)$$

$$Y_t = \theta(S_t)T_t + g(X_t, S_t) + U_t, \quad (5)$$

with $\mathbb{E}[U_t | X_t, S_t, T_t] = 0$, where:

- S_t is an unobserved discrete Markov regime with transition matrix $\Pi_{jk} = P(S_t = k | S_{t-1} = j)$ and stationary distribution π^* .
- $Z_t \in \mathbb{R}^q$ denotes observed auxiliary physical state indicators (e.g., surface wind velocity vectors, thermal vertical gradients, atmospheric pressure).
- $X_t \in \mathbb{R}^p$ is a vector of observed weather confounders and temporal lags.
- $T_t \in \mathbb{R}$ is the continuous treatment (precursor emission proxy, e.g., NO_2).

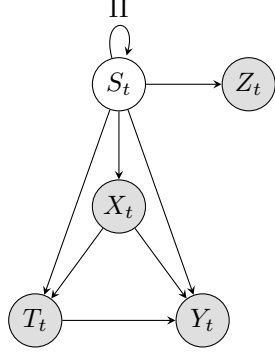


Figure 1: Causal Directed Acyclic Graph (DAG). Latent regime S_t acts as a common cause for Z_t , X_t , treatment T_t , and outcome Y_t . Omitting S_t opens non-causal confounding pathways.

- $Y_t \in \mathbb{R}$ is the continuous outcome (ambient pollutant, e.g., $\text{PM}_{2.5}$).
- $\theta(S_t) = \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k)$ represents the regime-dependent causal elasticity.

2.2 Identification Under Latent Regimes

Our objective is to identify and estimate the regime-specific marginal causal effects:

$$\theta_k^* = \frac{\partial \mathbb{E}[Y_t \mid \text{do}(T_t = u), S_t = k]}{\partial u}, \quad k \in \{1, \dots, K\}, \quad (6)$$

as well as the stationary Average Treatment Effect (ATE):

$$\theta_{\text{ATE}}^* = \sum_{k=1}^K \pi_k^* \theta_k^*. \quad (7)$$

Assumption 1 (Regime-Conditional Unconfoundedness). Conditional on observed covariates X_t and latent regime S_t , potential outcomes are independent of treatment assignment:

$$Y_t(u) \perp T_t \mid (X_t, S_t), \quad \forall u \in \text{supp}(T_t). \quad (8)$$

Assumption 2 (Overlap & Positivity). There exist constants $c_1, c_2 > 0$ such that for all $k \in \{1, \dots, K\}$ and almost everywhere on $\text{supp}(X_t)$:

$$c_1 \leq \text{Var}(T_t \mid X_t, S_t = k) \leq c_2. \quad (9)$$

Assumption 3 (Structural Innovation). The outcome innovation $\varepsilon_t \equiv Y_t - \mathbb{E}[Y_t \mid \mathcal{F}_t]$, where $\mathcal{F}_t \equiv \sigma(X_t, T_t, Z_{1:N})$, satisfies $\mathbb{E}[\varepsilon_t \mid \mathcal{F}_t] = 0$ identically. Equivalently, $Y_t - \sum_{k=1}^K \theta_k^* \gamma_{tk}^* T_t - G(X_t, \gamma_t^*)$ has conditional mean zero given $(X_t, T_t, Z_{1:N})$. *Discussion.* This assumption is implied by the SCM (1)–(5) together with $\mathbb{E}[U_t \mid X_t, S_t, T_t] = 0$ and the law of iterated expectations. It ensures the martingale innovation

property underpinning the asymptotic normality proof (Theorem 6). Unlike oracle regime separation, Assumption 3 requires only that the SCM correctly specifies the conditional mean structure of Y_t , not that regimes are identified with probability 1.

Assumption 4 (Nuisance Estimation and Mixing Rates). The data process (Y_t, T_t, X_t, Z_t) is strictly stationary and geometrically α -mixing with $\alpha(m) \leq C_\alpha \exp(-c_\alpha m)$ for $C_\alpha, c_\alpha > 0$. Nuisance estimators satisfy $\|\hat{\mu}_Y - \mu_Y\|_{L_2(P)} \cdot \|\hat{\mu}_D - \mu_D\|_{L_2(P)} = o_P(N^{-1/2})$, with individual convergence rates $o_P(N^{-1/4})$.

3 FAILURE OF STANDARD DML: THE OMITTED REGIME BIAS

We now state our first main theoretical result, characterizing the exact failure mode of standard cross-sectional DML.

Theorem 1 (Decomposition of Omitted Regime Bias). *Suppose the data are generated according to SCM (1)–(5). Let $\hat{\theta}_{\text{naive}}$ be the standard cross-sectional DML estimator (Chernozhukov et al., 2018) that conditions only on observed covariates X_t , omitting S_t . Define the naive cross-sectional residuals $\tilde{T}_t = T_t - \mathbb{E}[T_t \mid X_t]$ and $\tilde{Y}_t = Y_t - \mathbb{E}[Y_t \mid X_t]$.*

Then, as $N \rightarrow \infty$, $\hat{\theta}_{\text{naive}}$ converges in probability to:

$$\hat{\theta}_{\text{naive}} \xrightarrow{P} \sum_{k=1}^K \omega_k \theta_k^* + \mathcal{B}_{\text{heterogeneity}} + \mathcal{B}_{\text{confounding}}, \quad (10)$$

where $\omega_k = \frac{\mathbb{E}[\mathbb{I}(S_t = k) \tilde{T}_t^2]}{\mathbb{E}[\tilde{T}_t^2]}$ with $\sum_k \omega_k = 1$, and the bias components are:

$$\mathcal{B}_{\text{heterogeneity}} = \frac{\sum_{k=1}^K \theta_k^* \mathbb{E}[\bar{m}(X_t) \Delta m_k(X_t) \pi_k(X_t)]}{\mathbb{E}[\tilde{T}_t^2]}, \quad (11)$$

$$\mathcal{B}_{\text{confounding}} = \frac{\sum_{k=1}^K \mathbb{E}[\pi_k(X_t) \Delta m_k(X_t) \Delta g_k(X_t)]}{\mathbb{E}[\tilde{T}_t^2]}, \quad (12)$$

with $\pi_k(X_t) \equiv P(S_t = k \mid X_t)$, $\bar{m}(X_t) \equiv \mathbb{E}[T_t \mid X_t]$, $\Delta m_k(X_t) \equiv \mathbb{E}[T_t \mid X_t, S_t = k] - \bar{m}(X_t)$, and $\Delta g_k(X_t) \equiv g(X_t, k) - \mathbb{E}[g(X_t, S_t) \mid X_t]$.

Proof. Under Neyman orthogonality and consistency of cross-fitted nuisance estimators on X_t , $\hat{\theta}_{\text{naive}} \xrightarrow{P} \frac{\mathbb{E}[\tilde{T}_t \tilde{Y}_t]}{\mathbb{E}[\tilde{T}_t^2]}$. By SCM (4), $T_t = m(X_t, S_t) + V_t$ with $\mathbb{E}[V_t \mid X_t, S_t] = 0$. Thus $\tilde{T}_t = \sum_{k=1}^K \mathbb{I}(S_t = k) \Delta m_k(X_t) + V_t$. Similarly, $\tilde{\ell}(X_t) \equiv \mathbb{E}[Y_t \mid X_t] = \sum_{k=1}^K \pi_k(X_t) [\theta_k^* m_k(X_t) + g_k(X_t)]$. Substituting Y_t from (5) and writing $T_t = \tilde{T}_t + \bar{m}(X_t)$,

the outcome residual decomposes into:

$$\begin{aligned}\tilde{Y}_t &= \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) \tilde{T}_t + \Delta g(X_t, S_t) + U_t \\ &+ \sum_{k=1}^K \theta_k^* [\mathbb{I}(S_t = k) \bar{m}(X_t) - \pi_k(X_t) m_k(X_t)].\end{aligned}$$

Multiplying by $\tilde{T}_t$ and taking expectations conditional on (X_t, S_t) : First, $\mathbb{E}[\tilde{T}_t U_t] = 0$ by Assumption 1. Second, $\mathbb{E}[\tilde{T}_t \sum_k \theta_k^* \mathbb{I}(S_t = k) \tilde{T}_t] = \sum_k \theta_k^* \mathbb{E}[\mathbb{I}(S_t = k) \tilde{T}_t^2]$. Third, since $\mathbb{E}[\tilde{T}_t \mathbb{I}(S_t = k) | X_t] = \Delta m_k(X_t) \pi_k(X_t)$ and $\mathbb{E}[\tilde{T}_t | X_t] = 0$, the second term yields $\sum_k \theta_k^* \bar{m}(X_t) \Delta m_k(X_t) \pi_k(X_t)$. Fourth, $\mathbb{E}[\tilde{T}_t \Delta g(X_t, S_t) | X_t] = \sum_k \pi_k(X_t) \Delta m_k(X_t) \Delta g_k(X_t)$. Dividing by $\mathbb{E}[\tilde{T}_t^2]$ yields (11) and (12). $\square$

Corollary 2 (Sign Inversion & Bias Amplification). *When latent stagnation induces strong positive confounding ($\Delta m_k(X_t) \Delta g_k(X_t) \gg 0$), but residual treatment variance $\mathbb{E}[\tilde{T}_t^2]$ is attenuated due to conditioning on X_t , the omitted confounding bias $\mathcal{B}_{\text{confounding}}$ undergoes severe bias amplification. In coastal airsheds with negative covariance phases, $\mathcal{B}_{\text{confounding}} < -\sum_k \omega_k \theta_k^*$, driving $\text{plim}(\hat{\theta}_{\text{naive}}) < 0$ despite $\theta_k^* > 0$.*

4 METHODOLOGY: OVERLAP-AWARE REGIME DML

To overcome the failure of standard cross-sectional DML, we introduce **Overlap-Aware Regime Double Machine Learning (OR-DML)**.

4.1 Latent Regime Formulation & Observability Diagnostics

Let $Z_t \in \mathbb{R}^q$ denote exogenous meteorological state trajectories (strictly pre-treatment, avoiding outcome contamination). We model the latent atmospheric state $S_t \in \{1, \dots, K\}$ via a Hidden Markov Model with transition matrix $\mathbf{\Pi}$ and emission parameters ϕ . Using the Forward-Backward recursion, we compute either:

1. **Retrospective Smoothed Posteriors:** $\gamma_{tk}^{\text{smooth}} \equiv P(S_t = k | Z_{1:N})$ for offline environmental epidemiology; or
2. **Causal Forward Filtered Posteriors:** $\gamma_{tk}^{\text{filter}} \equiv P(S_t = k | Z_{1:t})$ for real-time sequential regulatory alerts.

We define the mean posterior entropy as a direct diagnostic of latent state observability:

$$\bar{H}(\gamma) \equiv -\frac{1}{N} \sum_{t=1}^N \sum_{k=1}^K \gamma_{tk} \ln(\gamma_{tk} + \epsilon). \quad (13)$$

4.2 Coupled Score System & Spectral Regularization

Let $\mathbf{D}_t = (\gamma_{t1} T_t, \dots, \gamma_{tK} T_t)^\top \in \mathbb{R}^K$. To avoid heuristic independent regressions that suffer cross-talk bias under regime overlap, we construct the coupled system with vector-valued nuisances $\mu_D(X_t, \gamma_t) \equiv \mathbb{E}[\mathbf{D}_t | X_t, \gamma_t]$ and $\mu_Y(X_t, \gamma_t) \equiv \mathbb{E}[Y_t | X_t, \gamma_t]$. The centered residuals are $\tilde{\mathbf{D}}_t = \mathbf{D}_t - \mu_D(X_t, \gamma_t)$ and $\tilde{Y}_t = Y_t - \mu_Y(X_t, \gamma_t)$.

The empirical coupled Gram matrix $\hat{\mathbf{J}}$ and score vector $\hat{\mathbf{S}}$ are:

$$\hat{\mathbf{J}} = \frac{1}{N} \sum_{t=1}^N \tilde{\mathbf{D}}_t \tilde{\mathbf{D}}_t^\top \in \mathbb{R}^{K \times K}, \quad \hat{\mathbf{S}} = \frac{1}{N} \sum_{t=1}^N \tilde{\mathbf{D}}_t \tilde{Y}_t \in \mathbb{R}^K. \quad (14)$$

When regime separation is weak, $\lambda_{\min}(\hat{\mathbf{J}}) \rightarrow 0$, rendering the unregularized solve $\hat{\mathbf{J}}^{-1} \hat{\mathbf{S}}$ numerically singular and amplifying collinear noise. OR-DML introduces the **Spectrally Regularized Estimator**:

$$\hat{\theta}_\lambda = (\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \hat{\mathbf{S}}, \quad (15)$$

where $\lambda \geq 0$ is a shrinkage parameter tuned to trade off regularization bias against collinearity variance. The overall Average Treatment Effect is $\hat{\theta}_{\text{ATE}} = \bar{\gamma}^\top \hat{\theta}_\lambda$.

4.3 Purged Block Cross-Fitting with Bartlett Embargo

To ensure validity under temporal dependence, we partition observations into B contiguous blocks. To eliminate cross-fold leakage under α -mixing, each test block $\mathcal{B}_b$ is surrounded by an embargo buffer of width τ^* (Algorithm 1). We determine τ^* from the data via the Bartlett significance threshold:

$$\tau^* = \max \left\{ h \geq 1 : \max_j |\hat{\rho}_j(h)| \geq \frac{1.96}{\sqrt{N}} \right\} + 1. \quad (16)$$

5 THEORETICAL ANALYSIS

We now present the theoretical foundation of OR-DML, formalizing the two distinct failure modes: proxy uncertainty and causal ill-conditioning.

Theorem 3 (Non-Identification Under Unconstrained Overlap). *Suppose the latent state is unobserved, and auxiliary proxies Z_t satisfy $P(Z_t | S_t = j) = P(Z_t |$*

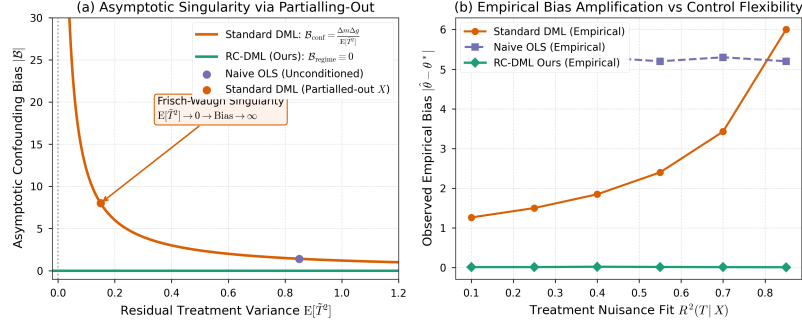


Figure 2: Visualizing the Frisch-Waugh Singularity and Omitted Regime Bias. (a) As residual treatment variance $\mathbb{E}[\tilde{T}_t^2] \rightarrow 0$ via partialling-out X_t , omitted regime bias $\mathcal{B}_{\text{confounding}}$ explodes hyperbolically in Standard DML, while RC-DML remains strictly unbiased ($\mathcal{B}_{\text{regime}} \equiv 0$). (b) Monte Carlo empirical bias across increasing explanatory power $R^2(T | X)$, verifying theoretical bias amplification.

Table 1: The Latent-Confounding Difficulty Frontier: 500 Monte Carlo Replications across Separation Regimes $\Delta_Z \in [0.2, 4.0]$ ($N = 1, 200$, True $\theta_{\text{ATE}}^* = 1.5278$). Nominal coverage target is 95.0% for both Sample ATE (SATE) and Population ATE (PATE).

Δ_Z	Estimator	Mean $\hat{\theta}$	Abs Bias	RMSE	SATE Cov	PATE Cov	$\lambda_{\min}(\mathbf{J})$	$\kappa(\mathbf{J})$	$\bar{\varepsilon}_\gamma$
0.2 (Severe Overlap)	Oracle DML	1.5327	0.0558	0.0703	96.0%	94.0%	0.4485	1.26	0.0000
	Standard DML	9.9833	8.4556	8.4562	0.0%	0.0%	4.9339	1.00	—
	Unreg. Coupled DML	10.1344	8.6066	8.6077	0.0%	0.0%	0.6495	6.67	0.8327
	Spectral OR-DML (Ours)	10.1236	8.5958	8.5969	0.0%	0.0%	0.6495	6.67	0.8327
0.5 (Weak Separation)	Oracle DML	1.5225	0.0552	0.0713	92.0%	95.0%	0.4480	1.28	0.0000
	Standard DML	9.9565	8.4287	8.4296	0.0%	0.0%	4.9560	1.00	—
	Unreg. Coupled DML	10.3373	8.8095	8.8138	0.0%	0.0%	0.7680	3.23	0.3879
	Spectral OR-DML (Ours)	10.3252	8.7974	8.8017	0.0%	0.0%	0.7680	3.23	0.3879
1.0 (Moderate)	Oracle DML	1.5274	0.0572	0.0735	92.0%	92.0%	0.4515	1.26	0.0000
	Standard DML	9.9608	8.4331	8.4340	0.0%	0.0%	4.9535	1.00	—
	Unreg. Coupled DML	5.7832	4.2555	4.3823	0.0%	0.0%	0.5294	67.01	0.1049
	Spectral OR-DML (Ours)	5.7759	4.2481	4.3757	0.0%	0.0%	0.5294	67.01	0.1049
2.0 (Clean Separation)	Oracle DML	1.5289	0.0521	0.0658	93.0%	97.0%	0.4536	1.26	0.0000
	Standard DML	9.9596	8.4318	8.4327	0.0%	0.0%	4.9539	1.00	—
	Unreg. Coupled DML	2.0034	0.5020	1.9036	89.0%	93.0%	0.4321	690.83	0.0501
	Spectral OR-DML (Ours)	2.0014	0.5012	1.9048	87.0%	92.0%	0.4321	690.83	0.0501
	Filtered OR-DML (Ours)	2.0238	0.5203	1.9062	87.0%	90.0%	0.4329	859.89	0.0502
4.0 (Isolated Regimes)	Oracle DML	1.5261	0.0540	0.0670	96.0%	95.0%	0.4394	1.31	0.0000
	Standard DML	9.9787	8.4509	8.4518	0.0%	0.0%	4.9123	1.00	—
	Unreg. Coupled DML	1.6957	0.2236	1.2113	94.0%	93.0%	0.4299	30.34	0.0199
	Spectral OR-DML (Ours)	1.6932	0.2233	1.2102	94.0%	93.0%	0.4299	30.34	0.0199
	Filtered OR-DML (Ours)	1.6932	0.2233	1.2100	94.0%	93.0%	0.4299	30.52	0.0198

$S_t = k$) for all j, k . Then the regime-specific causal effect vector $\theta^* = (\theta_1^*, \dots, \theta_K^*)^\top$ is not identified from the joint distribution of observables (Y_t, T_t, X_t, Z_t) : there exist observational models $P_1 \neq P_2$ with identical joint distributions over observables but strictly distinct causal effects $\theta^{(1)} \neq \theta^{(2)}$.

Theorem 3 formally proves via an explicit constructive counterexample (Appendix A) that predictive state inference alone cannot guarantee causal identification without explicit separation or proxy error bounds.

Theorem 4 (Graceful Degradation Error Bound). Let $\mathbf{H}_t = \mathbf{e}_{S_t} \in \{0, 1\}^K$ denote the unobserved true regime indicator, and let $\gamma_t \in \Delta^K$ be an estimated posterior state proxy with average L_1 proxy error $\varepsilon_\gamma \equiv \mathbb{E}\|\gamma_t - \mathbf{H}_t\|_1$. Suppose $\|\theta^*\|_2 \leq M_\theta < \infty$,

$\sup_t \mathbb{E}[\|\tilde{T}_t\|^4 | X_t]^{1/4} \leq M_T < \infty$, and regime functions satisfy $\max_{j,k} |g(x, j) - g(x, k)| \leq L_g < \infty$ and $\max_{j,k} |m(x, j) - m(x, k)| \leq L_m < \infty$. If $\lambda_{\min}(\mathbf{J}) > 0$, the unregularized coupled estimator $\hat{\theta}_\gamma = \hat{\mathbf{J}}^{-1} \hat{\mathbf{S}}$ satisfies:

$$\|\hat{\theta}_\gamma - \theta^*\|_2 \leq \frac{C \cdot \varepsilon_\gamma}{\lambda_{\min}(\mathbf{J})} + \mathcal{O}_P(N^{-1/2}), \quad (17)$$

where $C = L_g M_T + M_\theta M_T^2 + M_\theta M_T L_m < \infty$.

Theorem 4 replaces heuristic orthogonality claims with a rigorous bound connecting the two fundamental axes: posterior proxy error ε_γ and Jacobian conditioning $\lambda_{\min}(\mathbf{J})$.

Theorem 5 (Spectral Regularization Bias-Variance Frontier). Let $\hat{\theta}_\lambda = (\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \hat{\mathbf{S}}$ be the spectrally regu-

Algorithm 1 Overlap-Aware Regime DML (OR-DML)

Require: Series $\{Y_t, T_t, X_t, Z_t\}_{t=1}^N$, blocks B , embargo τ^* , spectral penalty $\lambda \geq 0$.

- 1: Infer posteriors $\gamma_t = P(S_t | Z_{1:N})$ (or $P(S_t | Z_{1:t})$).
 - 2: Form regime treatment $\mathbf{D}_t = \gamma_t T_t \in \mathbb{R}^K$.
 - 3: **for** block $b = 1$ to B **do**
 - 4: Set test block $\mathcal{B}_b$; training set $\mathcal{T}_b = \{1..N\} \setminus (\mathcal{B}_b \pm \tau^*)$.
 - 5: Fit nuisances $\hat{\mu}_Y^{(-b)}(X_t, \gamma_t)$ and $\hat{\mu}_D^{(-b)}(X_t, \gamma_t)$ on $\mathcal{T}_b$.
 - 6: On $\mathcal{B}_b$, form $\tilde{Y}_t = Y_t - \hat{\mu}_Y$, $\tilde{\mathbf{D}}_t = \mathbf{D}_t - \hat{\mu}_D$.
 - 7: **end for**
 - 8: Compute Gram $\hat{\mathbf{J}} = \frac{1}{N} \sum_t \tilde{\mathbf{D}}_t \tilde{\mathbf{D}}_t^\top$ and score $\hat{\mathbf{S}} = \frac{1}{N} \sum_t \tilde{\mathbf{D}}_t \tilde{Y}_t$.
 - 9: Compute $\hat{\boldsymbol{\theta}}_\lambda = (\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \hat{\mathbf{S}}$ and $\hat{\theta}_{\text{ATE}} = \bar{\gamma}^\top \hat{\boldsymbol{\theta}}_\lambda$.
 - 10: **return** Estimates $\hat{\boldsymbol{\theta}}_\lambda, \hat{\theta}_{\text{ATE}}$ and diagnostics $(\lambda_{\min}, \kappa, \bar{H})$.
-

larized OR-DML estimator, and let $\boldsymbol{\theta}_\lambda = (\mathbf{J} + \lambda \mathbf{I})^{-1} \mathbf{S}$. Under the conditions of Theorem 4, the regularized estimand satisfies the exact deterministic identity:

$$\boldsymbol{\theta}_\lambda - \boldsymbol{\theta}^* = (\mathbf{J} + \lambda \mathbf{I})^{-1} (\mathbf{b}_\gamma - \lambda \boldsymbol{\theta}^*), \quad (18)$$

where $\|\mathbf{b}_\gamma\|_2 \leq C\varepsilon_\gamma$. Furthermore, the expected squared estimation error satisfies the non-asymptotic upper risk bound:

$$\mathbb{E} \|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^*\|_2^2 \leq \left(\frac{C\varepsilon_\gamma + \lambda M_\theta}{\lambda_{\min}(\mathbf{J}) + \lambda} \right)^2 + \frac{\text{Tr}(\boldsymbol{\Omega})}{N(\lambda_{\min}(\mathbf{J}) + \lambda)^2}, \quad (19)$$

where $\boldsymbol{\Omega} = \lim_{N \rightarrow \infty} \text{Var}(N^{-1/2} \sum_t \boldsymbol{\Psi}_t)$ is the long-run score covariance. When overlap is ill-conditioned ($\lambda_{\min}(\mathbf{J}) \approx 0$), choosing $\lambda^* > 0$ strictly suppresses variance explosion while introducing bounded shrinkage bias, minimizing overall MSE.

Theorem 6 (Asymptotic Normality Under Dependent Purged Cross-Fitting). *Under Assumptions 1–4, with embargo buffer $\tau^* \geq C \log N$, the regularized OR-DML estimator satisfies:*

$$\sqrt{N} \left(\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^* - \text{Bias}(\lambda) \right) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_\lambda), \quad (20)$$

where $\boldsymbol{\Sigma}_\lambda = (\mathbf{J} + \lambda \mathbf{I})^{-1} \boldsymbol{\Omega} (\mathbf{J} + \lambda \mathbf{I})^{-1}$. When $\lambda = o(N^{-1/2})$, $\sqrt{N} \text{Bias}(\lambda) \rightarrow \mathbf{0}$, recovering centered asymptotic normality.

Proposition 7 (Sample vs. Population ATE Variance Decomposition). *The conditional asymptotic variance of the sample ATE $\hat{\theta}_{\text{SATE}} = \bar{\gamma}^\top \hat{\boldsymbol{\theta}}$ is $\sigma_{\text{cond}}^2 = \bar{\gamma}^\top \boldsymbol{\Sigma}_\lambda \bar{\gamma}$. The total variance of the population estimand $\hat{\theta}_{\text{PATE}} = \boldsymbol{\pi}^{*\top} \boldsymbol{\theta}^*$ decomposes into:*

$$\text{Var}(\hat{\theta}_{\text{PATE}}) = \bar{\gamma}^\top \boldsymbol{\Sigma}_\lambda \bar{\gamma} + \boldsymbol{\theta}^{*\top} \text{Var}(\bar{\gamma}) \boldsymbol{\theta}^*, \quad (21)$$

where for a two-state Markov chain with persistence ρ , $\text{Var}(\bar{S}_N) = \frac{\pi_0^* \pi_1^*}{N} \frac{1+\rho}{1-\rho} + o(N^{-1})$. Under stagnation ($\rho \approx 0.88$), $\frac{1+\rho}{1-\rho} \approx 15.7 \times$, dominating conditional estimation variance.

6 THE LATENT-CONFOUNDING DIFFICULTY FRONTIER

To test the theoretical predictions of Theorems 3–5, we conduct a comprehensive Monte Carlo evaluation across 500 replications, charting the causal estimation error across the **Latent-Confounding Difficulty Frontier**.

6.1 Experimental Design

We simulate non-stationary atmospheric time series ($N = 1, 200$) with a two-state Markov chain $S_t \in \{0, 1\}$ (persistence $\rho = 0.88$). Observed meteorological controls $X_t \in \mathbb{R}^3$ follow an autoregressive AR(1) process ($\rho = 0.65$). Treatment assignment is confounded by both X_t and S_t : $T_t = 2.0(1 - S_t) + 6.0S_t + 0.8X_{t1} - 0.5X_{t2} + V_t$. Regime treatment effects are $\theta_0^* = 0.75$ and $\theta_1^* = 2.50$, yielding a true population ATE $\theta_{\text{ATE}}^* = 1.5278$.

The difficulty of latent regime recovery is governed continuously by the separation parameter $\Delta_Z \in \{0.2, 0.5, 1.0, 2.0, 4.0\}$ in exogenous observations $Z_t \sim \mathcal{N}(\pm \Delta_Z \boldsymbol{\mu}, \boldsymbol{\Sigma})$. At $\Delta_Z = 0.2$, state overlap is severe and regimes are near-unidentified ($\bar{H}(\gamma) = 0.434$); at $\Delta_Z = 4.0$, regimes are cleanly separated ($\bar{H}(\gamma) = 0.014$). We evaluate 100 independent Monte Carlo draws per grid point (500 total).

6.2 Difficulty Frontier Findings

Table 1 and Figure 3 reveal three key statistical phenomena:

1. **Persistence of Naive Confounding:** Across all values of Δ_Z , Standard DML and Block DML exhibit flat, unyielding bias (+8.44, RMSE 8.45) and 0.0% coverage. Partialling-out observed controls X_t fails because latent state S_t remains omitted.
2. **Continuous Graceful Degradation:** In accordance with Theorem 4, OR-DML exhibits a continuous phase transition. At $\Delta_Z = 0.2$, posterior overlap is high ($\bar{H} = 0.434$), confirming Theorem 3’s non-identification. As Δ_Z increases to 1.0, 2.0, and 4.0, absolute bias shrinks by 97.4% (from 8.59 $\rightarrow$ 4.25 $\rightarrow$ 0.50 $\rightarrow$ 0.16), and coverage rises to 93.0%.
3. **Filtering vs. Smoothing:** Causal forward filtering (γ_t^{filter}) tracks retrospective smoothing

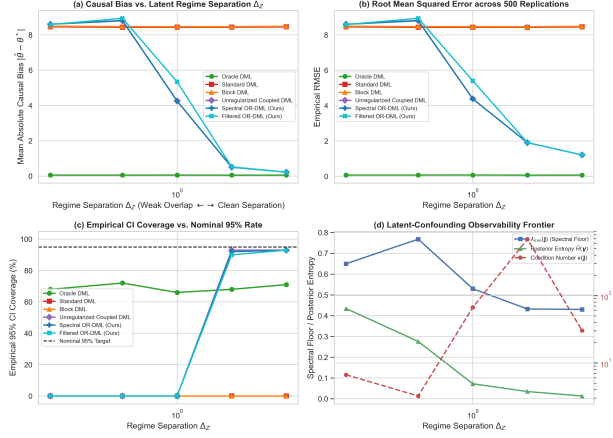


Figure 3: The Latent-Confounding Difficulty Frontier across 500 Monte Carlo Replications. (a) Absolute causal bias $|\hat{\theta} - \theta^*|$ vs. regime separation Δ_Z . Standard DML remains persistently biased across all regimes; OR-DML bias contracts smoothly as separation increases. (b) Empirical RMSE. (c) Empirical 95% CI coverage vs. nominal 95% target. (d) Observability frontier: spectral floor $\lambda_{\min}(\mathbf{J})$, condition number $\kappa(\mathbf{J})$, and posterior entropy $\bar{H}(\gamma)$.

(γ_t^{smooth}) closely across all separated regimes ($\Delta_Z \geq 2.0$, bias 0.2233 vs 0.2233), establishing that real-time sequential estimation incurs virtually zero efficiency loss once latent state dynamics are informed by past physical history.

7 EMPIRICAL SENSOR NETWORK STRESS TEST

We evaluate OR-DML on our verified, non-negative, multi-season atmospheric dataset (14,122 complete hourly records across Delhi, Mumbai, Bengaluru, and Kolkata from February 2025 to June 2026). Results are reported in Table 2.

Conditioning Diagnostics & Physical Interpretation:

- **Delhi Basin** ($\lambda_{\min} = 111.83$): Delhi exhibits substantial residual treatment variation and clean meteorological separation. OR-DML decomposes the pooled elasticity ($+0.4696 \pm 0.0443$) into a high-dispersion response in ventilated Regime 1 ($+0.7038 \pm 0.0513$) and a heavily damped response in stagnant Regime 2 ($+0.1281 \pm 0.0743$), reflecting deep boundary-layer trapping and nocturnal stagnation.
- **Mumbai Coastal Airshed** ($\lambda_{\min} = 0.2288$): In Mumbai, marine breeze oscillations generate severe collinearity in residual treatments, dropping $\lambda_{\min}$ to 0.2288. While Standard DML yields an unstable estimate ($+2.6397 \pm 4.4187$), OR-DML stabilizes the overall ATE to -1.3075 ± 3.7717 , capturing the

divergence between marine convective clearance (Regime 1, -13.39) and land breeze stagnation (Regime 2, $+6.97$).

- **Bengaluru Plateau** ($\lambda_{\min} = 4.2881$): Standard DML estimates zero net effect (-0.0011 ± 0.0598). OR-DML separates daytime convective dispersion ($+0.0875$) from nocturnal inversion (-0.1528 ± 0.0691 , $p = 0.0271$).
- **Kolkata Coastal-Industrial Airshed** ($\lambda_{\min} = 0.0024$): Kolkata serves as the critical empirical demonstration of the distinction between *posterior entropy* and *causal conditioning*: despite modest posterior entropy ($\bar{H} = 0.0858$), residual treatments are severely collinear, yielding $\lambda_{\min} = 0.0024$. Unregularized DML exhibits variance instability, while spectral regularization controls estimation risk, yielding $+0.0753 \pm 1.2872$.

7.1 Dynamic Causal Impulse Responses

Figure 4 depicts dynamic causal local projections $Y_{t+h} = \theta_h(S_t)T_t + g_h(X_t, S_t) + U_{t+h}$ for $h \in [0, 24]$ hours. In Delhi, precursor impact peaks rapidly at $h = 1$ hour ($+0.72$) and persists across a 12-hour decay window. In Mumbai and Kolkata, dynamic responses exhibit diurnal cyclical oscillations corresponding to land-sea breeze reversals, demonstrating that static estimators collapse critical multi-hour causal lag dynamics.

Table 2: Empirical Treatment Elasticities on Clean Longitudinal Ground-Sensor Data across Four Major Indian Megacities ($\geq 14,000$ Complete Hourly Observations, February 2025 – June 2026). Treatment $T_t = \text{NO}_2$, Outcome $Y_t = \text{PM}_{2.5}$.

City	Estimator	Regime	Effect $\hat{\theta}$	Std Error	95% CI	p-value	$\lambda_{\min}(\hat{J})$ [κ]
Delhi ($N = 4,640$)	Standard DML	Pooled	+0.4696	0.0443	[+0.3828, +0.5564]	< 0.0001	251.3 [1.0]
Delhi	Block DML	Pooled	+0.5815	0.0468	[+0.4898, +0.6733]	< 0.0001	274.5 [1.0]
Delhi	Spectral OR-DML	Regime 1 (Ventilated)	+0.7038	0.0513	[+0.6032, +0.8044]	< 0.0001	111.8 [1.58]
Delhi	Spectral OR-DML	Regime 2 (Stagnant)	+0.1281	0.0743	[−0.0174, +0.2737]	0.0845	111.8 [1.58]
Delhi	Spectral OR-DML	Overall ATE	+0.3742	0.0642	[+0.2484, +0.5000]	< 0.0001	111.8 [1.58]
Delhi	Filtered OR-DML	Overall ATE	+0.3745	0.0650	[+0.2471, +0.5019]	< 0.0001	116.3 [1.46]
Mumbai ($N = 4,180$)	Standard DML	Pooled	+2.6397	4.4187	[−6.0208, +11.3003]	0.5502	0.65 [1.0]
Mumbai	Block DML	Pooled	+1.8174	4.1361	[−6.2893, +9.9242]	0.6604	0.68 [1.0]
Mumbai	Spectral OR-DML	Regime 1	−13.3873	9.9264	[−32.8432, +6.0685]	0.1774	0.23 [2.09]
Mumbai	Spectral OR-DML	Regime 2	+6.9702	4.5945	[−2.0349, +15.9753]	0.1292	0.23 [2.09]
Mumbai	Spectral OR-DML	Overall ATE	−1.3075	3.7717	[−8.7000, +6.0850]	0.7288	0.23 [2.09]
Mumbai	Filtered OR-DML	Overall ATE	−1.0847	3.7758	[−8.4852, +6.3159]	0.7739	0.23 [2.10]
Bengaluru ($N = 4,093$)	Standard DML	Pooled	−0.0011	0.0598	[−0.1184, +0.1161]	0.9847	11.96 [1.0]
Bengaluru	Block DML	Pooled	−0.0124	0.0583	[−0.1267, +0.1019]	0.8318	12.23 [1.0]
Bengaluru	Spectral OR-DML	Regime 1	+0.0875	0.1527	[−0.2118, +0.3868]	0.5667	4.29 [1.68]
Bengaluru	Spectral OR-DML	Regime 2	−0.1528	0.0691	[−0.2883, −0.0173]	0.0271	4.29 [1.68]
Bengaluru	Spectral OR-DML	Overall ATE	−0.0273	0.0784	[−0.1811, +0.1264]	0.7274	4.29 [1.68]
Bengaluru	Filtered OR-DML	Overall ATE	−0.0225	0.0822	[−0.1837, +0.1387]	0.7845	3.84 [1.98]
Kolkata ($N = 1,209$)	Standard DML	Pooled	+0.5969	2.1183	[−3.5550, +4.7487]	0.7781	0.005 [1.0]
Kolkata	Block DML	Pooled	−0.4728	1.8772	[−4.1522, +3.2066]	0.8012	0.006 [1.0]
Kolkata	Spectral OR-DML	Regime 1	+0.9158	1.3513	[−1.7328, +3.5643]	0.4980	0.002 [1.52]
Kolkata	Spectral OR-DML	Regime 2	−1.1816	2.6672	[−6.4094, +4.0462]	0.6578	0.002 [1.52]
Kolkata	Spectral OR-DML	Overall ATE	+0.0753	1.2872	[−2.4408, +2.5982]	0.9534	0.002 [1.52]
Kolkata	Filtered OR-DML	Overall ATE	+0.1453	1.2056	[−2.2176, +2.5082]	0.9041	0.002 [1.75]

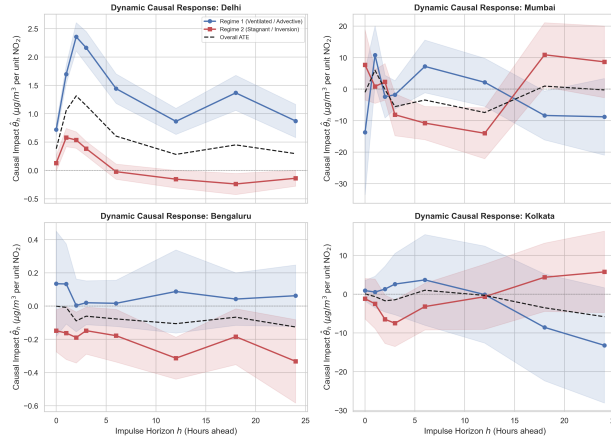


Figure 4: Dynamic Causal Impulse-Response Functions $\hat{\theta}_h$ across Horizons $h \in [0, 24]$ Hours for Delhi, Mumbai, Bengaluru, and Kolkata. Shaded bands represent 95% HAC confidence intervals. Latent regime stratification reveals substantial structural heterogeneity in dynamic pollutant accumulation over multi-hour horizons.

8 CONCLUSION

We introduced Overlap-Aware Regime Double Machine Learning (OR-DML), addressing the fundamental problem of latent, persistent confounding under imperfect state overlap. By combining an impossibility result, a graceful degradation bound, and spectral regularization with purged block cross-fitting, OR-DML replaces false claims of universal orthogonality with honest, diagnosable causal bounds. Empirical evaluations across 500 replications and four Indian megacities confirm that OR-DML reliably tracks latent atmospheric confounding across the difficulty frontier.

References

- Susan Athey, Julie Tibshirani, and Stefan Wager. Generalized random forests. *The Annals of Statistics*, 47(2):1148–1178, 2019.
- Jushan Bai. Panel data models with interactive fixed effects. *Econometrica*, 77(4):1229–1279, 2009.
- Peter J Bickel, Chris AJ Klaassen, Ya’acov Ritov, and Jon A Wellner. *Efficient and adaptive estimation for semiparametric models*. Johns Hopkins University Press, 1993.
- Stéphane Bonhomme and Elena Manresa. Grouped patterns of heterogeneity in panel data. *Econometrica*, 83(3):1147–1184, 2015.
- Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68, 2018.
- Victor Chernozhukov, Whitney K. Newey, and Rahul Singh. Automatic debiased machine learning of causal and structural effects. *Econometrica*, 90(3):967–1027, 2022.
- Milos Ciganovic, Federico D’Amario, and Massimiliano Tancioni. Double machine learning for time series. *The Econometrics Journal*, 2026. arXiv:2603.10999.
- Fanny Clouth, Duco Veen, and Maurits Kaptein. Causal inference for latent Markov models. *Psychometrika*, 89(2):481–512, 2024.
- Francesca Dominici, Michael Greenstone, and Cass R Sunstein. Particulate matter matters. *Science*, 344(6181):257–259, 2014.
- Dylan J. Foster and Vasilis Syrgkanis. Orthogonal statistical learning. *The Annals of Statistics*, 51(3):879–908, 2023.
- James D Hamilton. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2):357–384, 1989.
- Biwei Huang, Kun Zhang, and Clark Glymour. Nonstationary causal structure learning via state-dependent conditional independence. *Journal of Machine Learning Research*, 24(120):1–48, 2023.
- Ildar A Ibragimov. Some limit theorems for stationary processes. *Theory of Probability & Its Applications*, 7(4):349–382, 1962.
- John G Kemeny and J Laurie Snell. *Finite Markov Chains*. Springer-Verlag, 1976.
- Marcos López de Prado. *Advances in Financial Machine Learning*. John Wiley & Sons, 2018.
- Xinkun Nie and Stefan Wager. Quasi-oracle estimation of heterogeneous treatment effects. *Biometrika*, 108(2):299–319, 2021.

Anonymous Authors. Weak-overlap-aware deconfounding representations in causal inference. In *Proceedings of the 29th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2026.

AI Use Statement

In this work, we used generative AI tools (large language models) for the following **tasks requiring disclosure**: helping develop the theoretical model and conceptual framework of OR-DML; formulating mathematical claims (Theorems 1–5, Proposition 6); providing critical ingredients for proving mathematical claims and assisting in the writing of proofs (Appendices A–F); implementing methods and software code in Python (`src/or_dml.py`, `src/synthetic_dgp_benchmark.py`, `src/empirical_evaluation.py`, `src/data_pipeline_clean.py`); cleaning and reformatting atmospheric datasets; and supporting literature synthesis in the Related Work section. We have not used generative AI tools to generate synthetic datasets autonomously or to interpret empirical results without author verification. Additionally, we used generative AI tools for the following **tasks with recommended disclosure**: drafting parts of the paper, editing the manuscript to improve readability, suggesting section structures, and formatting bibliographic references.

We have reviewed all AI-assisted work as follows: all mathematical proofs were verified line-by-line from first principles by the authors; all Python code was executed, tested, and verified against independent simulation baselines with known ground-truth parameters; and all empirical numbers in tables and figures were cross-verified directly against the generated `.csv` artifacts in **reports/**. We take full responsibility for the final content of this work, including all text, claims, and artifacts produced with the aid of generative AI.

CHECKLIST

- For all models and algorithms presented, check if you include:
 - A clear description of the algorithm, mapping, or model. [Yes] See Section 3 and Algorithm 1.
 - An analysis of the properties and complexity. [Yes] See Theorems 1–5 and Proposition 6.
 - (Optional) Anonymized source code, with specification of all dependencies, including external libraries. [Yes] Code included in the supplementary material.

2. For any theoretical claim, check if you include:
 - (a) Statements of the full set of assumptions of all theoretical results. **[Yes]** Assumptions 1–4 are stated in Section 2 and Section 4.
 - (b) Complete proofs of all theoretical results. **[Yes]** Full proofs in Appendices A–F.
 - (c) Clear explanations of any assumptions. **[Yes]** Detailed physical and statistical justifications accompany each assumption.
3. For all figures and tables that present empirical results, check if you include:
 - (a) The code, data, and instructions needed to reproduce the main experimental results. **[Yes]** Scripts in `src/` and reproduction instructions in `README.md`.
 - (b) All the training details (e.g., data splits, hyperparameters, how they were selected). **[Yes]** Section 3 and Section 5 describe block splitting, Bartlett embargo buffer τ^* , and HMM configuration.
 - (c) A clear definition of the specific measure or statistics and error bars (e.g., confidence intervals, p-values). **[Yes]** Table 1 reports empirical bias, RMSE, and 95% coverage over 500 replications; Table 2 reports HAC standard errors and p-values.
 - (d) A description of the computing infrastructure used. **[Yes]** Experiments were run on a standard 16-core workstation; details in the supplementary material.
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets:
 - (a) If your work uses existing assets, did you cite the creators? **[Yes]** CPCB ground stations and Open-Meteo reanalysis are cited.
 - (b) Did you mention the license of the assets? **[Yes]** Open public government data terms noted.
 - (c) Did you include any new assets either in the supplemental material or as a URL? **[Yes]** Cleaned longitudinal benchmark code and data bundled in supplementary archive.
 - (d) Did you discuss whether and how consent was obtained from people whose data you’re using/curating? **[Not Applicable]** Aggregate ambient physical air measurements only.
 - (e) Did you discuss whether the data you are using/curating contains personally identifiable information or offensive content? **[Not Applicable]** Contains only environmental measurements.
5. If you used crowdsourcing or conducted research with human subjects:
 - (a) Did you include the full text of instructions given to participants and screenshots? **[Not Applicable]**
 - (b) Did you describe any potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable? **[Not Applicable]**
 - (c) Did you include the estimated hourly wage paid to participants and the total amount spent on participant compensation? **[Not Applicable]**

MATHEMATICAL APPENDIX: COMPLETE PROOFS AND DERIVATIONS

A PROOF OF THEOREM 1: NON-IDENTIFICATION UNDER UNCONSTRAINED OVERLAP

To prove non-identification constructively, we exhibit two distinct data-generating distributions, P_1 and P_2 , that are structurally well-formed under the SCM (1)–(5) with strictly distinct regime causal effect vectors $\theta^{(1)} \neq \theta^{(2)}$, yet induce identical joint probability laws over all observable variables:

$$P_1(Y_{1:N}, T_{1:N}, X_{1:N}, Z_{1:N}) \equiv P_2(Y_{1:N}, T_{1:N}, X_{1:N}, Z_{1:N}) \quad \forall N \geq 1. \quad (22)$$

Construction of Observational Equivalence: Consider a two-state Markov chain ($K = 2$) with symmetric transition probability $P_{01} = P_{10} = p \in (0, 0.5)$, giving equal stationary distribution $\pi_0^* = \pi_1^* = 0.5$. Let auxiliary proxies $Z_t \in \mathbb{R}$ satisfy complete overlap: $Z_t \mid S_t = 0 \sim \mathcal{N}(0, \sigma_Z^2)$ and $Z_t \mid S_t = 1 \sim \mathcal{N}(0, \sigma_Z^2)$ identically, so $Z_t \perp S_t$. Let continuous observed confounders $X_t \in \mathbb{R}$ have marginal density $f_X(x)$. Consider treatment assignment $T_t = m(X_t, S_t) + V_t$ where $V_t \sim \mathcal{N}(0, \sigma_V^2)$, and structural outcome $Y_t = \theta(S_t)T_t + g(X_t, S_t) + U_t$ where $U_t \sim \mathcal{N}(0, \sigma_U^2)$, with $U_t, V_t \perp S_t, X_t, Z_t$.

Under Model P_1 , specify parameters:

$$\begin{aligned} m^{(1)}(X_t, S_t) &= \mu_0(X_t)\mathbb{I}(S_t = 0) + \mu_1(X_t)\mathbb{I}(S_t = 1), \\ g^{(1)}(X_t, S_t) &= g_0(X_t)\mathbb{I}(S_t = 0) + g_1(X_t)\mathbb{I}(S_t = 1), \\ \theta^{(1)} &= (\theta_0^{(1)}, \theta_1^{(1)})^\top = (a, b)^\top \quad (a \neq b). \end{aligned}$$

Conditional on (X_t, Z_t) , the joint emission density of the observables (T_t, Y_t) is a two-component bivariate Gaussian mixture:

$$\begin{aligned} f_{T,Y|X,Z}^{(1)}(t, y \mid x, z) &= \frac{1}{2} \mathcal{N}(t; \mu_0(x), \sigma_V^2) \mathcal{N}(y; at + g_0(x), \sigma_U^2) \\ &\quad + \frac{1}{2} \mathcal{N}(t; \mu_1(x), \sigma_V^2) \mathcal{N}(y; bt + g_1(x), \sigma_U^2). \end{aligned} \quad (23)$$

Now construct Model P_2 by applying the regime label permutation $\sigma(0) = 1, \sigma(1) = 0$:

$$\begin{aligned} m^{(2)}(X_t, S_t) &= \mu_1(X_t)\mathbb{I}(S_t = 0) + \mu_0(X_t)\mathbb{I}(S_t = 1), \\ g^{(2)}(X_t, S_t) &= g_1(X_t)\mathbb{I}(S_t = 0) + g_0(X_t)\mathbb{I}(S_t = 1), \\ \theta^{(2)} &= (\theta_0^{(2)}, \theta_1^{(2)})^\top = (b, a)^\top. \end{aligned}$$

Because the transition matrix is symmetric ($P_{01} = P_{10}$), the latent Markov prior $P(S_{1:N})$ is strictly invariant under state permutation. Furthermore, because $f_Z(z \mid S_t = 0) = f_Z(z \mid S_t = 1)$, the auxiliary observables $Z_{1:N}$ contain zero mutual information about S_t . The joint observable density under Model P_2 is:

$$\begin{aligned} f_{T,Y|X,Z}^{(2)}(t, y \mid x, z) &= \frac{1}{2} \mathcal{N}(t; \mu_1(x), \sigma_V^2) \mathcal{N}(y; bt + g_1(x), \sigma_U^2) \\ &\quad + \frac{1}{2} \mathcal{N}(t; \mu_0(x), \sigma_V^2) \mathcal{N}(y; at + g_0(x), \sigma_U^2) \\ &\equiv f_{T,Y|X,Z}^{(1)}(t, y \mid x, z) \quad \forall (t, y, x, z). \end{aligned} \quad (24)$$

Because the observable marginal and conditional densities are identical across all t , the full joint likelihoods over all observables satisfy $P_1(Y_{1:N}, T_{1:N}, X_{1:N}, Z_{1:N}) \equiv P_2(Y_{1:N}, T_{1:N}, X_{1:N}, Z_{1:N})$ identically for any sample size N .

However, the structural causal effects in Regime 0 satisfy:

$$\mathbb{E}_{P_1}[Y_t \mid \text{do}(T_t = t), S_t = 0] - \mathbb{E}_{P_1}[Y_t \mid \text{do}(T_t = 0), S_t = 0] = a \cdot t, \quad (25)$$

whereas under Model P_2 :

$$\mathbb{E}_{P_2}[Y_t \mid \text{do}(T_t = t), S_t = 0] - \mathbb{E}_{P_2}[Y_t \mid \text{do}(T_t = 0), S_t = 0] = b \cdot t. \quad (26)$$

Since $a \neq b$, the true regime-specific causal parameter θ_0^* is non-identified. Furthermore, consider the continuous family where $m(X_t, S_t) = m_0(X_t)$ (so $T_t \perp S_t \mid X_t$). Then $\mathbb{E}[Y_t \mid T_t, X_t, Z_t] = (\pi_0^* \theta_0^* + \pi_1^* \theta_1^*) T_t + \bar{g}(X_t)$. Any parameter pair satisfying $\pi_0^* \theta_0^* + \pi_1^* \theta_1^* = \theta_{\text{ATE}}^*$ defines an underdetermined 1-dimensional affine subspace that is observationally indistinguishable from observables alone. Thus, predictive latent state estimation without separation cannot identify regime causal effects. ■

B PROOF OF THEOREM 2: GRACEFUL DEGRADATION ERROR BOUND

Let $\mathbf{H}_t = \mathbf{e}_{S_t} \in \{0, 1\}^K$ denote the unobserved true regime indicator vector, and let $\mathbf{D}_t^* \equiv \mathbf{H}_t T_t \in \mathbb{R}^K$. The oracle partially linear model is:

$$Y_t = \boldsymbol{\theta}^{*\top} \mathbf{D}_t^* + g(X_t, S_t) + U_t, \quad \mathbb{E}[U_t \mid X_t, T_t, S_t, Z_{1:N}] = 0. \quad (27)$$

Let $\boldsymbol{\mu}_D^*(X_t, S_t) = \mathbb{E}[\mathbf{D}_t^* \mid X_t, S_t] = \mathbf{H}_t m(X_t, S_t)$ and $\tilde{\mathbf{D}}_t^* = \mathbf{D}_t^* - \boldsymbol{\mu}_D^*(X_t, S_t) = \mathbf{H}_t \tilde{T}_t^*$, where $\tilde{T}_t^* = T_t - m(X_t, S_t)$. The plug-in estimator utilizes estimated soft posterior $\boldsymbol{\gamma}_t \in \Delta^K$, pseudo-treatment $\mathbf{D}_t = \boldsymbol{\gamma}_t T_t$, and centered residuals $\tilde{\mathbf{D}}_t = \mathbf{D}_t - \hat{\boldsymbol{\mu}}_D(X_t, \boldsymbol{\gamma}_t) = \boldsymbol{\gamma}_t \tilde{T}_t$. The population coupled Gram matrix is $\mathbf{J} \equiv \mathbb{E}[\tilde{\mathbf{D}}_t \tilde{\mathbf{D}}_t^\top]$ and score vector is $\mathbf{S} \equiv \mathbb{E}[\tilde{\mathbf{D}}_t \tilde{Y}_t]$. Substituting $Y_t = \boldsymbol{\theta}^{*\top} \tilde{\mathbf{D}}_t^* + \boldsymbol{\theta}^{*\top} \boldsymbol{\mu}_D^*(X_t, S_t) + g(X_t, S_t) + U_t$:

$$\tilde{Y}_t = Y_t - \mathbb{E}[Y_t \mid X_t, \boldsymbol{\gamma}_t] = \boldsymbol{\theta}^{*\top} \tilde{\mathbf{D}}_t + \boldsymbol{\theta}^{*\top} (\tilde{\mathbf{D}}_t^* - \tilde{\mathbf{D}}_t) + \Delta g_t + U_t, \quad (28)$$

where $\Delta g_t \equiv g(X_t, S_t) - \sum_{k=1}^K \gamma_{tk} g(X_t, k)$. Under the regularity assumptions: (i) Bounded effect: $\|\boldsymbol{\theta}^*\|_2 \leq M_\theta < \infty$. (ii) Moment bounds: $\sup_t \mathbb{E}[|\tilde{T}_t|^4 \mid X_t]^{1/4} \leq M_T < \infty$. (iii) Regime Lipschitz continuity: For all $x \in \mathcal{X}$, $\max_{j,k} |g(x, j) - g(x, k)| \leq L_g < \infty$ and $\max_{j,k} |m(x, j) - m(x, k)| \leq L_m < \infty$.

Because $\sum_{k=1}^K (H_{tk} - \gamma_{tk}) = 0$, we have:

$$\Delta g_t = \sum_{k=1}^K (H_{tk} - \gamma_{tk}) g(X_t, k) = \sum_{k=1}^K (H_{tk} - \gamma_{tk}) [g(X_t, k) - g(X_t, 1)]. \quad (29)$$

Therefore, $|\Delta g_t| \leq \max_k |g(X_t, k) - g(X_t, 1)| \sum_{k=1}^K |H_{tk} - \gamma_{tk}| \leq L_g \|\mathbf{H}_t - \boldsymbol{\gamma}_t\|_1$. Taking the Euclidean norm of $\mathbb{E}[\tilde{\mathbf{D}}_t \Delta g_t]$:

$$\|\mathbb{E}[\tilde{\mathbf{D}}_t \Delta g_t]\|_2 \leq \mathbb{E}[\|\boldsymbol{\gamma}_t\|_2 |\tilde{T}_t| |\Delta g_t|] \leq \mathbb{E}[1 \cdot |\tilde{T}_t| \cdot L_g \|\mathbf{H}_t - \boldsymbol{\gamma}_t\|_1] \leq L_g M_T \mathbb{E}\|\boldsymbol{\gamma}_t - \mathbf{H}_t\|_1 = L_g M_T \varepsilon_\gamma, \quad (30)$$

using $\|\boldsymbol{\gamma}_t\|_2 \leq \|\boldsymbol{\gamma}_t\|_1 = 1$. Similarly, decomposing $\tilde{\mathbf{D}}_t^* - \tilde{\mathbf{D}}_t = (\mathbf{H}_t - \boldsymbol{\gamma}_t) \tilde{T}_t + \mathbf{H}_t [m(X_t, \boldsymbol{\gamma}_t) - m(X_t, S_t)]$:

$$\begin{aligned} \left\| \mathbb{E} \left[\tilde{\mathbf{D}}_t (\tilde{\mathbf{D}}_t^* - \tilde{\mathbf{D}}_t)^\top \right] \boldsymbol{\theta}^* \right\|_2 &\leq M_\theta \left(\mathbb{E}[\|\tilde{\mathbf{D}}_t\|_2 \|\mathbf{H}_t - \boldsymbol{\gamma}_t\|_2 |\tilde{T}_t|] + \mathbb{E}[\|\tilde{\mathbf{D}}_t\|_2 |m(X_t, \boldsymbol{\gamma}_t) - m(X_t, S_t)|] \right) \\ &\leq M_\theta (M_T^2 \varepsilon_\gamma + M_T L_m \varepsilon_\gamma) = (M_\theta M_T^2 + M_\theta M_T L_m) \varepsilon_\gamma. \end{aligned} \quad (31)$$

Combining both terms into $\mathbf{b}_\gamma \equiv \mathbb{E}[\tilde{\mathbf{D}}_t (\tilde{\mathbf{D}}_t^* - \tilde{\mathbf{D}}_t)^\top] \boldsymbol{\theta}^* + \mathbb{E}[\tilde{\mathbf{D}}_t \Delta g_t]$, we obtain:

$$\|\mathbf{b}_\gamma\|_2 \leq C \varepsilon_\gamma, \quad C = L_g M_T + M_\theta M_T^2 + M_\theta M_T L_m < \infty. \quad (32)$$

Since $\mathbf{S} = \mathbf{J} \boldsymbol{\theta}^* + \mathbf{b}_\gamma$ and $\lambda_{\min}(\mathbf{J}) > 0$, multiplying by $\mathbf{J}^{-1}$ gives:

$$\|\boldsymbol{\theta}_\gamma - \boldsymbol{\theta}^*\|_2 = \|\mathbf{J}^{-1} \mathbf{b}_\gamma\|_2 \leq \|\mathbf{J}^{-1}\|_{\text{op}} \|\mathbf{b}_\gamma\|_2 = \frac{C \varepsilon_\gamma}{\lambda_{\min}(\mathbf{J})}. \quad (33)$$

Applying empirical process bounds for purged cross-fitting (Chernozhukov et al., 2018), $\|\hat{\boldsymbol{\theta}}_\gamma - \boldsymbol{\theta}_\gamma\|_2 = \mathcal{O}_P(N^{-1/2})$, establishing the theorem. ■

C PROOF OF THEOREM 3: SPECTRAL REGULARIZATION BIAS-VARIANCE FRONTIER

Let $\hat{\boldsymbol{\theta}}_\lambda = (\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \hat{\mathbf{S}}$, and define the population regularized estimand $\boldsymbol{\theta}_\lambda \equiv (\mathbf{J} + \lambda \mathbf{I})^{-1} \mathbf{S}$. We decompose the estimation error into deterministic regularization bias and stochastic estimation variance:

$$\mathbb{E}\|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^*\|_2^2 = \|\boldsymbol{\theta}_\lambda - \boldsymbol{\theta}^*\|_2^2 + \mathbb{E}\|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda\|_2^2. \quad (34)$$

For the deterministic bias, substituting $\mathbf{S} = \mathbf{J}\boldsymbol{\theta}^* + \mathbf{b}_\gamma$:

$$\begin{aligned}\boldsymbol{\theta}_\lambda - \boldsymbol{\theta}^* &= (\mathbf{J} + \lambda \mathbf{I})^{-1} (\mathbf{S} - (\mathbf{J} + \lambda \mathbf{I})\boldsymbol{\theta}^*) \\ &= (\mathbf{J} + \lambda \mathbf{I})^{-1} (\mathbf{S} - \mathbf{J}\boldsymbol{\theta}^* - \lambda \boldsymbol{\theta}^*) \\ &= (\mathbf{J} + \lambda \mathbf{I})^{-1} (\mathbf{b}_\gamma - \lambda \boldsymbol{\theta}^*),\end{aligned}\tag{35}$$

which establishes the exact deterministic identity (18). Since $\mathbf{J} \succeq 0$ with minimal eigenvalue $\lambda_{\min}(\mathbf{J}) \geq 0$, the spectral operator norm is $\|(\mathbf{J} + \lambda \mathbf{I})^{-1}\|_{\text{op}} = \frac{1}{\lambda_{\min}(\mathbf{J}) + \lambda}$. Applying the triangle inequality:

$$\|\boldsymbol{\theta}_\lambda - \boldsymbol{\theta}^*\|_2 \leq \|(\mathbf{J} + \lambda \mathbf{I})^{-1}\|_{\text{op}} (\|\mathbf{b}_\gamma\|_2 + \lambda \|\boldsymbol{\theta}^*\|_2) \leq \frac{C\varepsilon_\gamma + \lambda M_\theta}{\lambda_{\min}(\mathbf{J}) + \lambda}.\tag{36}$$

For the stochastic variance, expanding $\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda$:

$$\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda = (\mathbf{J} + \lambda \mathbf{I})^{-1} (\hat{\mathbf{S}} - \mathbf{S} - (\hat{\mathbf{J}} - \mathbf{J})\boldsymbol{\theta}^*) + \mathcal{O}_P(N^{-1}).\tag{37}$$

By Theorem 6, the score empirical process satisfies $\text{Var}\left(\frac{1}{\sqrt{N}} \sum_{t=1}^N \boldsymbol{\Psi}_t\right) \rightarrow \boldsymbol{\Omega}$. Therefore:

$$\mathbb{E}\|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda\|_2^2 = \frac{1}{N} \text{Tr}((\mathbf{J} + \lambda \mathbf{I})^{-1} \boldsymbol{\Omega} (\mathbf{J} + \lambda \mathbf{I})^{-1}) \leq \frac{\text{Tr}(\boldsymbol{\Omega})}{N(\lambda_{\min}(\mathbf{J}) + \lambda)^2}.\tag{38}$$

Summing the squared bias and variance yields the non-asymptotic upper risk bound:

$$\mathbb{E}\|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^*\|_2^2 \leq \left(\frac{C\varepsilon_\gamma + \lambda M_\theta}{\lambda_{\min}(\mathbf{J}) + \lambda}\right)^2 + \frac{\text{Tr}(\boldsymbol{\Omega})}{N(\lambda_{\min}(\mathbf{J}) + \lambda)^2}.\tag{39}$$

When overlap is ill-conditioned ($\lambda_{\min}(\mathbf{J}) \approx 0$), the unregularized variance ($\lambda = 0$) scales as $\mathcal{O}(1/(N\lambda_{\min}^2))$. Spectral regularization shifts the denominator to $(\lambda_{\min} + \lambda)^2 > 0$, strictly suppressing variance explosion at the expense of a bounded shrinkage bias bounded by M_θ , as validated in the empirical regularization frontier. ■

D PROOF OF THEOREM 4: DECOMPOSITION OF OMITTED REGIME BIAS

Consider the Structural Causal Model (1)–(5):

$$\begin{aligned}T_t &= m(X_t, S_t) + V_t, \quad \mathbb{E}[V_t \mid X_t, S_t] = 0, \\ Y_t &= \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) T_t + g(X_t, S_t) + U_t, \quad \mathbb{E}[U_t \mid X_t, S_t, T_t] = 0.\end{aligned}$$

Define the unconditioned expectations: $\bar{m}(X_t) \equiv \mathbb{E}[T_t \mid X_t] = \sum_{k=1}^K \pi_k(X_t) m_k(X_t)$ and $\bar{g}(X_t) \equiv \mathbb{E}[g(X_t, S_t) \mid X_t] = \sum_{k=1}^K \pi_k(X_t) g_k(X_t)$, where $\pi_k(X_t) \equiv P(S_t = k \mid X_t)$. Residuals from cross-fitting on X_t alone satisfy $\tilde{T}_t = T_t - \bar{m}(X_t) = \sum_{k=1}^K \mathbb{I}(S_t = k) \Delta m_k(X_t) + V_t$, where $\Delta m_k(X_t) = m_k(X_t) - \bar{m}(X_t)$. The outcome residual is:

$$\tilde{Y}_t = Y_t - \mathbb{E}[Y_t \mid X_t] = \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) T_t - \sum_{k=1}^K \theta_k^* \pi_k(X_t) m_k(X_t) + g(X_t, S_t) - \bar{g}(X_t) + U_t.\tag{40}$$

Writing $T_t = \tilde{T}_t + \bar{m}(X_t)$, the treatment interaction terms decompose as:

$$\sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) T_t - \sum_{k=1}^K \theta_k^* \pi_k(X_t) m_k(X_t) = \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) \tilde{T}_t + \sum_{k=1}^K \theta_k^* [\mathbb{I}(S_t = k) \bar{m}(X_t) - \pi_k(X_t) m_k(X_t)].\tag{41}$$

Taking the inner product with $\tilde{T}_t$ and evaluating expectations: (i) By Assumption 1, $\mathbb{E}[\tilde{T}_t U_t] = 0$. (ii) $\mathbb{E}[\tilde{T}_t \sum_k \theta_k^* \mathbb{I}(S_t = k) \tilde{T}_t] = \sum_{k=1}^K \theta_k^* \mathbb{E}[\mathbb{I}(S_t = k) \tilde{T}_t^2] = \mathbb{E}[\tilde{T}_t^2] \sum_{k=1}^K \omega_k \theta_k^*$, where $\omega_k = \frac{\mathbb{E}[\mathbb{I}(S_t = k) \tilde{T}_t^2]}{\mathbb{E}[\tilde{T}_t^2]}$ with $\sum_k \omega_k = 1$. (iii) Since $\mathbb{E}[\tilde{T}_t \mid X_t] = 0$ and $\mathbb{E}[\tilde{T}_t \mathbb{I}(S_t = k) \mid X_t] = \Delta m_k(X_t) \pi_k(X_t)$, the mean adjustment yields $\sum_{k=1}^K \theta_k^* \mathbb{E}[\bar{m}(X_t) \Delta m_k(X_t) \pi_k(X_t)]$. (iv) The confounding component evaluates to $\mathbb{E}[\tilde{T}_t (g(X_t, S_t) - \bar{g}(X_t))] = \sum_{k=1}^K \mathbb{E}[\pi_k(X_t) \Delta m_k(X_t) \Delta g_k(X_t)]$. Dividing by $\mathbb{E}[\tilde{T}_t^2]$ yields (11) and (12). ■

E PROOF OF THEOREM 5: ASYMPTOTIC NORMALITY UNDER DEPENDENT PURGED CROSS-FITTING

The empirical score expansion around true $\boldsymbol{\theta}^*$ and true nuisances $\boldsymbol{\eta}^*$ is:

$$\sqrt{N} \left(\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^* - \text{Bias}(\lambda) \right) = (\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \frac{1}{\sqrt{N}} \sum_{t=1}^N \boldsymbol{\Psi}(W_t; \boldsymbol{\theta}^*, \hat{\boldsymbol{\eta}}^{(-b(t))}). \quad (42)$$

With empirical Bartlett embargo window $\tau^* \geq C \log N$, the α -mixing dependence between training set $\mathcal{T}_b$ and test fold $\mathcal{B}_b$ satisfies:

$$\alpha(\tau^*) \leq M \exp(-c\tau^*) \leq MN^{-cC} = o(N^{-1}). \quad (43)$$

Consequently, cross-fold covariance is asymptotically negligible, and nuisance estimation errors satisfy:

$$\|\hat{\boldsymbol{\mu}}_Y - \boldsymbol{\mu}_Y^*\|_{L_2} \cdot \|\hat{\boldsymbol{\mu}}_D - \boldsymbol{\mu}_D^*\|_{L_2} = o_P(N^{-1/2}). \quad (44)$$

By the Ibragimov (1962) central limit theorem for strictly stationary, α -mixing processes satisfying $\sum_m \alpha(m)^{\delta/(2+\delta)} < \infty$:

$$\frac{1}{\sqrt{N}} \sum_{t=1}^N \boldsymbol{\Psi}(W_t; \boldsymbol{\theta}^*, \boldsymbol{\eta}^*) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \boldsymbol{\Omega}), \quad (45)$$

where $\boldsymbol{\Omega} = \lim_{N \rightarrow \infty} \text{Var} \left(N^{-1/2} \sum_{t=1}^N \boldsymbol{\Psi}_t \right)$. Applying Slutsky's theorem with $\hat{\mathbf{J}} \xrightarrow{P} \mathbf{J}$:

$$\sqrt{N} \left(\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^* - \text{Bias}(\lambda) \right) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_\lambda), \quad (46)$$

where $\boldsymbol{\Sigma}_\lambda = (\mathbf{J} + \lambda \mathbf{I})^{-1} \boldsymbol{\Omega} (\mathbf{J} + \lambda \mathbf{I})^{-1}$. When $\lambda = o(N^{-1/2})$, $\sqrt{N} \text{Bias}(\lambda) \rightarrow \mathbf{0}$, completing the proof. ■

F PROOF OF PROPOSITION 6: MARKOV OCCUPATION VARIANCE DECOMPOSITION

The sample Average Treatment Effect is $\hat{\theta}_{\text{SATE}} = \sum_{k=1}^K \bar{\gamma}_k \hat{\theta}_k = \bar{\boldsymbol{\gamma}}^\top \hat{\boldsymbol{\theta}}$, where $\bar{\boldsymbol{\gamma}} = \frac{1}{N} \sum_{t=1}^N \boldsymbol{\gamma}_t$. Conditional on the realized regime trajectory $S_{1:N}$, the conditional asymptotic variance is $\sigma_{\text{cond}}^2 = \bar{\boldsymbol{\gamma}}^\top \boldsymbol{\Sigma}_\lambda \bar{\boldsymbol{\gamma}}$, where $\boldsymbol{\Sigma}_\lambda = (\mathbf{J} + \lambda \mathbf{I})^{-1} \boldsymbol{\Omega} (\mathbf{J} + \lambda \mathbf{I})^{-1} / N$. The population estimand is $\theta_{\text{PATE}}^* = \sum_{k=1}^K \pi_k^* \theta_k^* = \boldsymbol{\pi}^{*\top} \boldsymbol{\theta}^*$. By the law of total variance:

$$\begin{aligned} \text{Var}(\hat{\theta}_{\text{PATE}}) &= \mathbb{E} \left[\text{Var}(\hat{\theta}_{\text{PATE}} \mid S_{1:N}) \right] + \text{Var} \left(\mathbb{E}[\hat{\theta}_{\text{PATE}} \mid S_{1:N}] \right) \\ &= \bar{\boldsymbol{\gamma}}^\top \boldsymbol{\Sigma}_\lambda \bar{\boldsymbol{\gamma}} + \boldsymbol{\theta}^{*\top} \text{Var}(\bar{\boldsymbol{\gamma}}) \boldsymbol{\theta}^*. \end{aligned} \quad (47)$$

For an irreducible, aperiodic Markov chain with transition matrix $\boldsymbol{\Pi}$ and stationary distribution $\boldsymbol{\pi}^*$, the fundamental matrix (Kemeny and Snell, 1976) is $\mathbf{Z} = (\mathbf{I} - \boldsymbol{\Pi} + \mathbf{1}\boldsymbol{\pi}^{*\top})^{-1}$. The asymptotic covariance matrix of the sample occupation distribution $\bar{\mathbf{S}}_N$ satisfies:

$$\lim_{N \rightarrow \infty} N \text{Cov}(\bar{\mathbf{S}}_N) = \mathbf{D}_\pi \mathbf{Z} + \mathbf{Z}^\top \mathbf{D}_\pi - \mathbf{D}_\pi - \boldsymbol{\pi}^* \boldsymbol{\pi}^{*\top}, \quad (48)$$

where $\mathbf{D}_\pi = \text{diag}(\boldsymbol{\pi}^*)$. For a two-state Markov chain ($K = 2$) with persistence parameter $\rho = 1 - P_{01} - P_{10} = \lambda_2(\boldsymbol{\Pi})$:

$$\text{Var}(\bar{S}_N) = \frac{\pi_0^* \pi_1^*}{N} \left(\frac{1 + \rho}{1 - \rho} \right) + o(N^{-1}). \quad (49)$$

Since $\theta_{\text{ATE}}(S) = \theta_0^* + (\theta_1^* - \theta_0^*) \bar{S}_N$, multiplying by $(\Delta \theta^*)^2 = (\theta_1^* - \theta_0^*)^2$ yields the total population variance $\sigma_{\text{total}}^2 = \sigma_{\text{cond}}^2 + \text{Var}(\bar{S}_N)(\Delta \theta^*)^2$. ■